# KMS Enterprise Knowledge Platform: Capability Areas and Differentiators

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| **Product** | KMS (Knowledge Management System) |
| **Document** | Capability areas and differentiators, Enterprise Knowledge Platform |
| **Version** | 1.0 (draft), 2026-10-19 |
| **Source** | `docs/ekp/KMS_EKP_PRODUCT_REQUIREMENTS_SPECIFICATION.md` §3–§4 (areas follow NYPA RFP Q26-7821RJM §2.1 headings) |

This document summarizes each EKP capability area from the Product Requirements Specification, and the approach KMS takes in that area that sets it apart from a standard search or document-management product. Where an area has no distinctive approach yet (mostly areas still marked Gap), the table says so.

Status counts are taken from PRS §3 (as of 2026-10-08).

## Capability Areas

| Area | Description | Unique approach |
|---|---|---|
| **Unified Access, Search and Navigation** (E1–E4) | A single, customizable landing page; one search across KMS uploads and connected repositories (SharePoint mandatory, Content Server and ServiceNow targeted) that respects source permissions; semantic/hybrid search; faceted navigation on the taxonomy. *Status: 4 Partial.* | **Hybrid vector + keyword search in one ranking.** Every document is chunked and embedded (OpenAI `text-embedding-3-small`, 1536-dim) into **pgvector**; queries run cosine-distance vector search alongside **BM25** keyword ranking, merged into a single hybrid score (65% semantic / 35% keyword). **LLM query expansion** rewrites each query into alternative phrasings before retrieval, and **knowledge-graph expansion** pulls in related documents. Permission and department scope are applied *inside* the SQL candidate query, so restricted content is never ranked, not just hidden afterwards. |
| **Personalization & Role-Aware Experience** (E5–E6) | Role- and department-scoped discovery, personal homepages, topic hubs, saved searches and watchlists; recommendations and trending content in the user's domain. *Status: 1 Met, 1 Partial.* | **Embedding-based recommendations with a feedback loop.** Seed queries from the user's activity and topics are embedded and matched against the vector index, then re-ranked by learned per-user feedback (helpful +0.08, not helpful −0.12, capped at ±0.3), with a plain-language reason shown for each recommendation. **Topic hubs are built on auto-discovered topics** (see topic modeling below), so they exist from day one without a hand-built taxonomy. **Expert discovery** ranks people per topic from capped, time-decayed signals (90-day half-life) weighted toward real engagement (questions asked, lessons authored, citations) over clicks, with an itemized "why this expert" breakdown. |

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|---|---|---|
| **Conversational Query & Answering** (E7–E10) | Natural-language answers beside search results; retrieval-augmented, cited, permission-aware answers; follow-up questions and refinement; confidence scores and explainability. *Status: 3 Met, 1 Partial.* | **Permission-filtered RAG over the hybrid index.** Retrieval is RBAC-filtered *before* generation, so the model never sees content the user can't access; every answer carries typed citations (document, decision, lesson) and denied attempts are logged to `ChatAuditLog`. **Retrieval-strength confidence:** the high/medium/low score is computed from the vector distances of the top retrieved chunks (with genuine keyword hits credited as strong matches), not from the model's own self-assessment. Answers also draw on the knowledge graph, extracted decisions and lessons learned, and can synthesize across projects. |
| **Knowledge Content Coverage** (E11–E14) | Multiple asset formats (documents, PPT, PDF, video, images); site pages, wikis and knowledge articles; indexing, previews and thumbnails; configurable asset templates. *Status: 2 Partial, 2 Gap.* | **Dependency-free OCR for scanned PDFs** (pdf.js + tesseract.js, no native binaries), so scanned content enters the same vector index and is citable like born-digital text. Spreadsheets are extracted into searchable text. Otherwise no distinctive approach yet: PPTX, images, video/audio transcription, site pages and templates are still to be built. |
| **Content Enrichment and Semantic Layer Integration** (E15–E17) | Managed metadata with hierarchical terms and synonyms; integration with NYPA's taxonomy/ontology tool; knowledge-graph-enriched search, navigation and conversational querying. *Status: 1 Met, 2 Gap.* | **Unsupervised topic modeling.** Each document's embedding is matched to topic **centroid embeddings** (cosine similarity, with a small Bhattacharyya keyword-distribution term); documents join the closest topic (≥ 0.60) or start a new one, and centroids and keyword distributions update incrementally as documents are added or removed. Topics are named by an LLM from the document's subject. **Topic evolution tracking** captures periodic snapshots (document volume, expert count, vocabulary drift) so trends survive even after a topic is retired. **Organizational knowledge graph** links documents, topics, people, projects, decisions and lessons; relationship confidence blends embedding similarity (0.46), topic overlap (0.18), entity overlap (0.16), project context (0.12) and lesson evidence (0.08), and edges are typed (references, supersedes, shares topic/entity, etc.). Planned: auto-discovered topics offered to admins as candidate managed terms (E15.6). |
| **Content Ingestion** (E18) | Easy drag-and-drop or file-picker upload, several files at once, with progress and errors shown. *Status: Met.* | **Ingest-time AI pipeline.** One upload triggers extraction/OCR, chunking and embedding, summarization, auto-classification, topic assignment, duplicate and conflict checks, and decision/entity extraction in a background worker, so a document is searchable, tagged and linked without manual steps. |
| **Automated Metadata Generation** (E19–E21) | AI-generated description with human review; auto-classification and auto-tagging with human review; minimum metadata completeness before publishing. *Status: 1 Met, 2 Partial.* | **Confidence-gated human-in-the-loop classification.** The LLM suggests category and department with confidence scores; anything below threshold goes to a Needs Review queue instead of being applied silently, and classification re-runs automatically when a file is replaced or a synced file changes. Every document also gets an AI summary, key points and a reading guide at ingest. |

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|---|---|---|
| **Governance and Content Lifecycle Management** (E22–E27) | Configurable publishing workflows; decommission (archive, retire, supersede, delete) with audit trail; ownership and stewardship; review/validation workflows; stale content detection; duplicate and near-duplicate detection. *Status: 1 Met, 4 Partial, 1 Gap.* | **Vector-based duplicate and contradiction detection.** Exact duplicates are caught by content hash; near-duplicates by **document-embedding similarity ≥ 0.92**. Beyond duplicates, a **conflict detector** finds semantically close document pairs via vector search and has an LLM verify whether they actually contradict each other (recorded only at ≥ 0.7 confidence), surfacing conflicting guidance for review. Publishing is gated on a reviewer (department admin or org admin), never the author alone. Configurable workflows, review policies and automated stale-content triggers are still to be built. |
| **Analytics and Dashboards** (E28–E31) | Content health (recency, ownership, review status, broken links, duplicates); usage and adoption (searches, CTR, time-to-answer, top queries, bounce rate); search quality (no-results queries, precision/recall proxies, dwell time); governance compliance. *Status: 3 Partial, 1 Gap.* | **Knowledge-gap detection from the answer engine.** Questions that produce zero citations or low-confidence answers are flagged as knowledge gaps, showing where content is missing rather than only how often people search. A health panel surfaces answer confidence and open conflicts, and dwell time is captured per document view. Full usage, search-quality and governance dashboards are still to be built. |
| **Integration and Authoring** (E32) | M365 as the single source of truth: Microsoft (Entra ID) sign-in, M365 kept as the master copy, “Open in SharePoint/Office” links, edits made in M365 and picked up on sync. *Status: Partial.* | **Microsoft Graph delta sync** brings in only changed SharePoint/OneDrive items, with per-run sync history. **Email-first Microsoft sign-in** routes users to Entra ID SSO based on their email domain, and a “Connect Microsoft” admin flow uses Microsoft’s admin-consent endpoint. Synced content flows through the same embedding, topic and graph pipeline as uploads, so SharePoint documents get the same semantic search and enrichment. |

## Cross-cutting differentiators

| Technique | Where it’s used |
|---|---|
| **Vector search** (pgvector, 1536-dim embeddings, cosine distance) | Hybrid search (E3), RAG answers (E8), answer confidence (E10), recommendations (E6), topic assignment (E15–E17), relationship discovery (E17), near-duplicate and conflict detection (E27) |
| **Topic modeling** (incremental centroid clustering, LLM-named topics, snapshot-based evolution tracking) | Topic hubs (E5), expert discovery (E5), knowledge graph (E17), candidate managed terms (E15.6, planned), relationship scoring (E17) |
| **Knowledge graph** (typed, confidence-scored edges across documents, topics, people, projects, decisions, lessons) | Search expansion (E3), graph explorer navigation (E17), graph-aware answers (E9, E17) |
| **Permission-first retrieval** (RBAC applied in the SQL query, before ranking or generation) | Every area: search, answers, citations, recommendations, graph, topic trends (PRS §2 universal rule, NFR-1) |

| Technique | Where it’s used |
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| **Human-in-the-loop AI** (confidence thresholds route uncertain AI output to people) | Classification (E20), conflicts (E27), knowledge gaps (E30); planned for descriptions (E19) |